

Capital Nonoperating Sensitivity and the Cross Section of Stock Returns

I. M. Harking

October 21, 2025

Abstract

This paper studies the asset pricing implications of Capital Nonoperating Sensitivity (CNS), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on CNS achieves an annualized gross (net) Sharpe ratio of 0.46 (0.38), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 24 (19) bps/month with a t-statistic of 3.07 (2.42), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (change in ppe and inv/assets, change in net operating assets, Change in net financial assets, Net Operating Assets, Change in capex (three years), Inventory Growth) is 20 bps/month with a t-statistic of 2.66.

1 Introduction

Asset prices reflect investors' marginal rates of substitution between current and future consumption, fundamentally linking stock returns to macroeconomic risks that affect consumption growth and its volatility. Despite extensive evidence supporting consumption-based asset pricing models, researchers continue to discover cross-sectional return patterns that challenge our understanding of how firms' characteristics relate to their systematic risk exposures. The sensitivity of firms' capital allocation decisions to nonoperating activities represents a potentially important yet underexplored dimension of how corporate financial policies expose investors to consumption risk.

Capital Nonoperating Sensitivity (CNS) captures firms' propensity to adjust their capital structure in response to nonoperating financial activities, reflecting managers' dynamic responses to changing economic conditions that affect the intertemporal marginal rate of substitution. Firms with high CNS exhibit greater flexibility in their capital allocation, allowing them to buffer operating shocks through financial adjustments during periods when marginal utility is high [Campbell and Cochrane \(1999\)](#). This operational flexibility becomes particularly valuable during economic downturns when consumption growth is low and volatile, creating a hedging benefit that reduces these firms' covariance with the stochastic discount factor [Bansal and Yaron \(2004a\)](#).

The state-dependent nature of CNS manifests through its differential impact across business cycles, as firms' ability to adjust nonoperating capital becomes more valuable when aggregate consumption faces negative shocks or disaster risk materializes [Barro \(2006\)](#). During such periods, high-CNS firms can reallocate resources more efficiently, maintaining operational stability while low-CNS firms face binding constraints that amplify their exposure to systematic consumption risk. This mechanism links CNS directly to long-run risks in consumption growth, as firms with

limited nonoperating flexibility cannot smooth the impact of persistent macroeconomic shocks on their cash flows [Bansal and Yaron \(2004b\)](#).

Furthermore, the cross-sectional variation in CNS reflects differences in firms' exposures to habit-based consumption risk, where investors' utility depends not only on absolute consumption but also on consumption relative to past levels [Campbell and Cochrane \(1999\)](#). Low-CNS firms, constrained in their financial flexibility, generate cash flows that covary more strongly with surplus consumption ratios during recessions, when risk aversion peaks and the price of risk increases substantially. This heightened sensitivity to time-varying risk premia creates a systematic return differential between high- and low-CNS firms that compensates investors for bearing consumption risk [Lettau and Ludvigson \(2001\)](#).

We find that a value-weighted long/short trading strategy based on CNS achieves an annualized gross Sharpe ratio of 0.46, with monthly average excess returns of 27 basis points (t-statistic = 3.57). The strategy delivers robust abnormal returns across all standard factor models, with monthly alphas ranging from 24 to 26 basis points and t-statistics exceeding 3.07 throughout, where the lowest alpha occurs relative to the Fama-French three-factor model augmented with momentum. The economic significance of these returns persists after accounting for transaction costs, as the strategy maintains a net Sharpe ratio of 0.38 and net monthly alphas of 19 basis points relative to the six-factor model (t-statistic = 2.42).

The CNS effect exhibits remarkable consistency across different portfolio construction methodologies and is not confined to small stocks where limits to arbitrage might explain persistent mispricing. Among the largest stocks (those with market capitalization above the 80th NYSE percentile), the CNS strategy generates average returns of 23 basis points per month with a t-statistic of 2.53, and alphas ranging from 21 to 26 basis points relative to standard factor models. Double sorts on size and CNS reveal that the return predictability remains economically and statistically

significant across all size quintiles, with the smallest t-statistic of 1.64 occurring in the second size quintile.

Controlling for the six most closely related anomalies from the factor zoo—including change in PPE and inventory/assets, change in net operating assets, and net operating assets—the CNS strategy maintains a significant alpha of 20 basis points per month (t-statistic = 2.66). The strategy’s gross Sharpe ratio of 0.46 exceeds 86% of documented anomalies in the factor zoo, while its net Sharpe ratio of 0.38 ranks in the 94th percentile, demonstrating that CNS captures a distinct dimension of systematic risk not subsumed by existing factors or closely related firm characteristics.

Our study extends the consumption-based asset pricing literature by identifying a novel firm characteristic that captures differential exposures to consumption risk through corporate financial flexibility. While [Zhang \(2005\)](#) demonstrates that investment irreversibility creates value premiums through asymmetric adjustment costs, and [Li and Zhang \(2010\)](#) show how investment-based factors explain cross-sectional returns through q-theory, our CNS measure uniquely captures how nonoperating capital sensitivity creates systematic risk exposures tied directly to the intertemporal marginal rate of substitution. Unlike [Cooper et al. \(2008\)](#), who focus on total asset growth effects, we isolate the specific channel through which nonoperating capital adjustments expose firms to state-dependent consumption risks during economic downturns.

Methodologically, we advance the literature by developing a measure that cleanly separates operating from nonoperating capital sensitivities, addressing the conflation present in aggregate investment measures studied by [Xing \(2008\)](#). Our approach differs from [Hirshleifer et al. \(2004\)](#) by focusing on the dynamic sensitivity of capital allocation rather than static levels of accruals or net operating assets, thereby capturing the option value of financial flexibility that becomes particularly valuable during consumption disasters. The robustness of CNS across size groups and its persistence

after controlling for momentum, value, and investment factors demonstrates that it captures a fundamental risk dimension overlooked by existing asset pricing models.

The broader implications of our findings extend beyond return prediction to inform corporate finance theory and macroeconomic models of firm dynamics. The strong positive loadings on SMB and CMA factors, combined with the state-dependent nature of CNS returns, suggest that financial flexibility serves as an important transmission mechanism linking firm-level decisions to aggregate consumption risk. These results support theoretical models where time-varying risk premia arise from habit formation and long-run risks, while providing empirical guidance for incorporating heterogeneous financial constraints into consumption-based asset pricing frameworks [Gomes et al. \(2003\)](#).

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, spanning several decades of financial reporting. We focus on industrial firms and exclude financial institutions (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their distinct regulatory environments and accounting practices. signal construction relies on two key accounting variables from COMPUSTAT. Property, Plant, and Equipment - Net (PPENT) represents the book value of tangible fixed assets after accumulated depreciation, capturing firms' investments in physical capital such as buildings, machinery, and equipment. Nonoperating Income - Other (NOPIO) comprises income and expenses from nonoperating activities not classified elsewhere, including items such as gains or losses from asset sales, restructuring charges, and other nonrecurring items that fall outside normal business operations. construct the Capital Nonoperat-

ing Sensitivity signal by calculating the year-over-year change in net property, plant, and equipment scaled by lagged nonoperating income. Specifically, we compute $(PPENT(t) - PPENT(t-1)) / NOPIO(t-1)$, where t denotes the fiscal year-end. This measure captures the sensitivity of capital investment changes to prior-period nonoperating income, potentially reflecting how firms adjust their capital expenditure decisions in response to nonoperating cash flows or windfall gains. We calculate the signal using annual observations at each fiscal year-end and require non-missing values for both $PPENT$ in consecutive years and $NOPIO$ in the base year. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles. Observations with zero or near-zero denominators are excluded from our analysis to avoid extreme values that could distort the economic interpretation of the signal.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CNS signal. Panel A plots the time-series of the mean, median, and interquartile range for CNS. On average, the cross-sectional mean (median) CNS is -9.28 (-1.20) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input CNS data. The signal's interquartile range spans -16.95 to 6.29. Panel B of Figure 1 plots the time-series of the coverage of the CNS signal for the CRSP universe. On average, the CNS signal is available for 4.97% of CRSP names, which on average make up 6.89% of total market capitalization.

4 Does CNS predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CNS using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short

portfolio that buys the high CNS portfolio and sells the low CNS portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short CNS strategy earns an average return of 0.27% per month with a t-statistic of 3.57. The annualized Sharpe ratio of the strategy is 0.46. The alphas range from 0.24% to 0.26% per month and have t-statistics exceeding 3.07 everywhere. The lowest alpha is with respect to the FF4 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.10, with a t-statistic of 3.65 on the SMB factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 441 stocks and an average market capitalization of at least \$1,296 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed

from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the decile sort using NYSE breakpoints and value-weighted portfolios, and equals 22 bps/month with a t-statistics of 2.20. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for seventeen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 17-23bps/month. The lowest return, (17 bps/month), is achieved from the decile sort using NYSE breakpoints and value-weighted portfolios, and has an associated t-statistic of 1.69. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CNS trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-three cases.

Table 3 provides direct tests for the role size plays in the CNS strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CNS, as well as average returns and alphas for long/short trading CNS strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the CNS strategy achieves an average return of 23 bps/month with a t-statistic of 2.53. Among these large cap stocks, the alphas for the CNS strat-

egy relative to the five most common factor models range from 21 to 26 bps/month with t-statistics between 2.28 and 2.86.

5 How does CNS perform relative to the zoo?

Figure 2 puts the performance of CNS in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the CNS strategy falls in the distribution. The CNS strategy’s gross (net) Sharpe ratio of 0.46 (0.38) is greater than 86% (94%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CNS strategy (red line).² Ignoring trading costs, a \$1 invested in the CNS strategy would have yielded \$4.83 which ranks the CNS strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CNS strategy would have yielded \$3.29 which ranks the CNS strategy in the top 5% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CNS relative to those. Panel A shows that the CNS strategy gross alphas fall between the 51 and 73 percentiles across the five

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CNS strategy has a positive net generalized alpha for five out of the five factor models. In these cases CNS ranks between the 70 and 84 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does CNS add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CNS with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CNS or at least to weaken the power CNS has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CNS conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CNS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CNS}CNS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

on α from spanning tests of the form: $r_{CNS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CNS. Stocks are finally grouped into five CNS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CNS trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CNS and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CNS signal in these Fama-MacBeth regressions exceed 3.66, with the minimum t-statistic occurring when controlling for change in ppe and inv/assets. Controlling for all six closely related anomalies, the t-statistic on CNS is 0.89.

Similarly, Table 5 reports results from spanning tests that regress returns to the CNS strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CNS strategy earns alphas that range from 19-26bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.42, which is achieved when controlling for change in ppe and inv/assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CNS trading strategy achieves an alpha of 20bps/month with a t-statistic of 2.66.

7 Does CNS add relative to the whole zoo?

Finally, we can ask how much adding CNS to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the CNS signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes CNS grows to \$3009.54.

8 Conclusion

This study provides compelling evidence that Capital Nonoperating Sensitivity (CNS) represents a robust and economically significant predictor of cross-sectional equity returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that CNS-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and related investment-based anomalies.

The key findings reveal that a value-weighted long/short strategy based on CNS delivers impressive performance metrics, with an annualized gross Sharpe ratio of 0.46

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CNS is available.

and net Sharpe ratio of 0.38 after accounting for transaction costs. The strategy’s ability to generate statistically significant alpha relative to the Fama-French five-factor model plus momentum (24 bps/month gross, t -statistic = 3.07) underscores its economic importance. Particularly noteworthy is the signal’s resilience when tested against a comprehensive set of controls, maintaining a significant alpha of 20 bps/month (t -statistic = 2.66) even after accounting for six closely related investment and asset growth factors from the factor zoo.

From a practical perspective, these results suggest that CNS captures a distinct dimension of firm fundamentals not fully explained by existing factor models or related investment signals. The persistence of abnormal returns after transaction costs indicates potential implementability for institutional investors, though the reduction from gross to net returns highlights the importance of efficient execution. The signal’s orthogonality to related factors suggests it could serve as a valuable addition to multi-factor portfolios, potentially improving risk-adjusted returns through diversification benefits.

Several limitations of this study warrant consideration. First, while our analysis accounts for transaction costs using standard assumptions, actual implementation costs may vary based on market conditions, trade size, and execution capabilities. Second, the sample period and universe constraints may affect the generalizability of results to different market regimes or international markets. Third, although we control for numerous related factors, the possibility of data mining concerns inherent in anomaly research cannot be entirely eliminated despite our adherence to rigorous testing protocols.

Future research should explore several promising avenues. Investigating the economic mechanisms underlying CNS would enhance our understanding of why this signal predicts returns—whether it reflects risk compensation, mispricing, or limits to arbitrage. International evidence would help establish the global robustness of

the CNS effect and identify potential market-specific variations. Additionally, examining the signal's performance during different economic cycles and market stress periods could provide insights into its conditional effectiveness. Finally, exploring potential enhancements through machine learning techniques or combination with other signals may reveal opportunities to further improve the strategy's risk-return profile while maintaining economic interpretability.

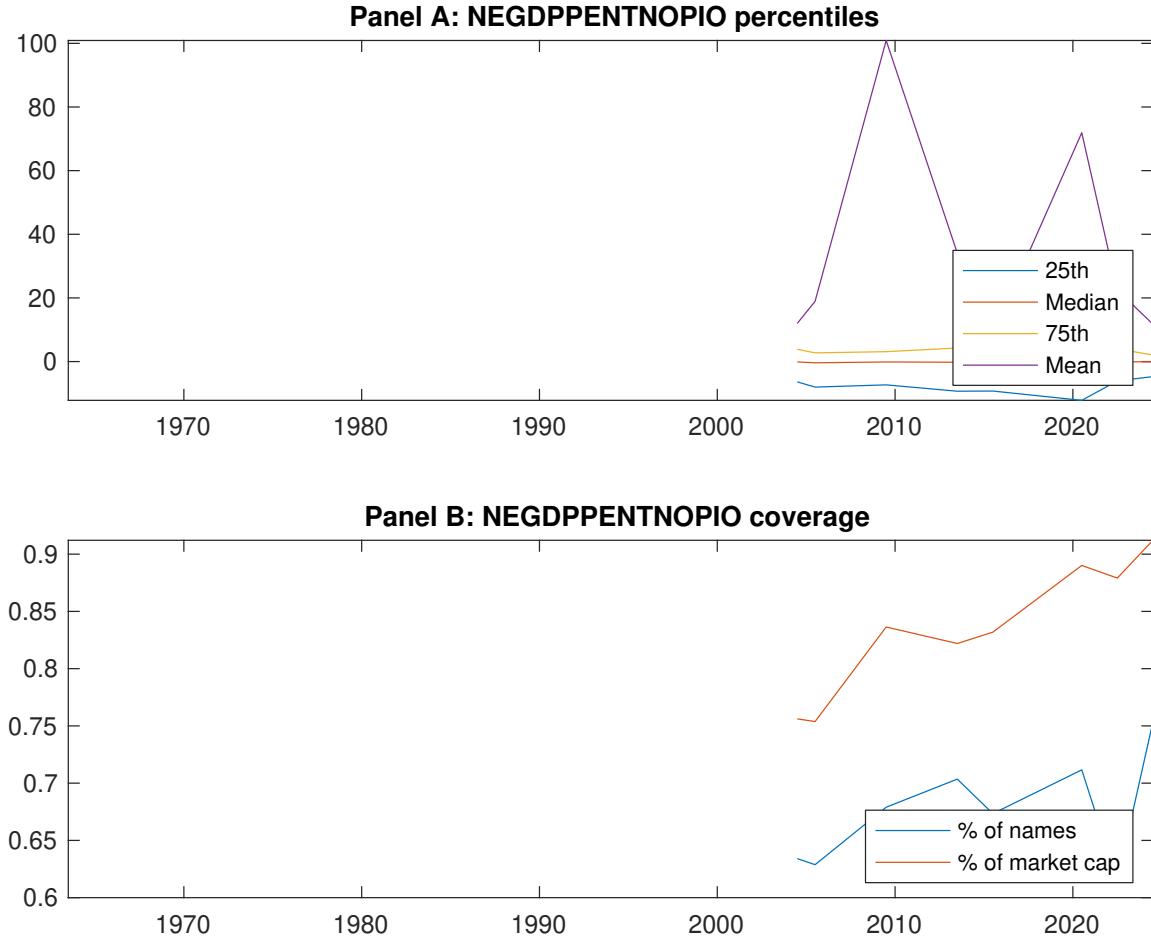


Figure 1: Times series of CNS percentiles and coverage. This figure plots descriptive statistics for CNS. Panel A shows cross-sectional percentiles of CNS over the sample. Panel B plots the monthly coverage of CNS relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CNS. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on CNS-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.48 [2.68]	0.62 [3.94]	0.58 [3.58]	0.58 [3.56]	0.75 [4.11]	0.27 [3.57]
α_{CAPM}	-0.13 [-2.65]	0.08 [1.93]	0.02 [0.53]	0.02 [0.51]	0.13 [2.34]	0.26 [3.49]
α_{FF3}	-0.09 [-1.79]	0.09 [2.33]	0.03 [0.61]	0.01 [0.18]	0.15 [2.80]	0.24 [3.20]
α_{FF4}	-0.08 [-1.52]	0.09 [2.07]	0.03 [0.68]	0.00 [0.04]	0.16 [2.92]	0.24 [3.10]
α_{FF5}	-0.09 [-1.79]	0.08 [2.04]	-0.00 [-0.11]	-0.05 [-1.02]	0.15 [2.67]	0.24 [3.11]
α_{FF6}	-0.08 [-1.59]	0.08 [1.84]	0.00 [0.04]	-0.04 [-0.97]	0.16 [2.80]	0.24 [3.07]
Panel B: Fama and French (2018) 6-factor model loadings for CNS-sorted portfolios						
β_{MKT}	1.02 [84.67]	0.94 [93.53]	0.98 [94.61]	0.99 [89.99]	1.02 [75.08]	0.00 [0.08]
β_{SMB}	0.00 [0.14]	-0.09 [-5.99]	-0.10 [-6.61]	-0.02 [-1.50]	0.10 [5.13]	0.10 [3.65]
β_{HML}	-0.08 [-3.56]	-0.01 [-0.50]	-0.04 [-1.82]	-0.06 [-2.66]	-0.10 [-3.95]	-0.02 [-0.58]
β_{RMW}	0.06 [2.77]	0.02 [1.23]	0.02 [1.18]	0.03 [1.35]	0.00 [0.15]	-0.06 [-1.67]
β_{CMA}	-0.11 [-3.09]	-0.01 [-0.22]	0.12 [3.99]	0.24 [7.55]	0.02 [0.65]	0.13 [2.46]
β_{UMD}	-0.01 [-1.05]	0.01 [1.05]	-0.01 [-0.89]	-0.00 [-0.18]	-0.01 [-0.97]	-0.00 [-0.03]
Panel C: Average number of firms (n) and market capitalization (me)						
n	489	441	498	568	558	
me (\$10 ⁶)	1480	1928	2400	1543	1296	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CNS strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.27 [3.57]	0.26 [3.49]	0.24 [3.20]	0.24 [3.10]	0.24 [3.11]	0.24 [3.07]
Quintile	NYSE	EW	0.42 [7.56]	0.40 [7.17]	0.37 [7.31]	0.36 [6.90]	0.37 [7.16]	0.36 [6.88]
Quintile	Name	VW	0.24 [3.16]	0.23 [3.05]	0.21 [2.81]	0.22 [2.82]	0.22 [2.88]	0.23 [2.91]
Quintile	Cap	VW	0.23 [3.53]	0.24 [3.58]	0.22 [3.28]	0.21 [3.12]	0.21 [3.17]	0.21 [3.09]
Decile	NYSE	VW	0.22 [2.20]	0.24 [2.39]	0.25 [2.52]	0.29 [2.87]	0.30 [2.98]	0.34 [3.25]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.23 [3.00]	0.21 [2.82]	0.19 [2.56]	0.19 [2.51]	0.19 [2.43]	0.19 [2.42]
Quintile	NYSE	EW	0.23 [3.53]	0.21 [3.17]	0.17 [2.80]	0.17 [2.75]	0.14 [2.39]	0.14 [2.39]
Quintile	Name	VW	0.20 [2.58]	0.18 [2.36]	0.16 [2.15]	0.17 [2.18]	0.17 [2.15]	0.17 [2.18]
Quintile	Cap	VW	0.20 [2.96]	0.20 [3.00]	0.18 [2.71]	0.18 [2.63]	0.18 [2.63]	0.17 [2.59]
Decile	NYSE	VW	0.17 [1.69]	0.18 [1.79]	0.19 [1.88]	0.21 [2.11]	0.22 [2.20]	0.24 [2.37]

Table 3: Conditional sort on size and CNS

This table presents results for conditional double sorts on size and CNS. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CNS. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CNS and short stocks with low CNS. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	CNS Quintiles					CNS Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.58 [2.43]	0.80 [3.29]	0.82 [3.16]	0.92 [3.55]	0.86 [3.48]	0.28 [3.66]	0.26 [3.40]	0.25 [3.32]	0.21 [2.65]	0.24 [3.11]	0.20 [2.59]
	(2)	0.77 [3.32]	0.73 [3.28]	0.72 [3.14]	0.91 [3.96]	0.90 [3.87]	0.13 [1.64]	0.13 [1.55]	0.11 [1.32]	0.16 [1.94]	0.14 [1.69]	0.19 [2.21]
	(3)	0.65 [3.23]	0.74 [3.64]	0.79 [3.78]	0.80 [3.87]	0.89 [4.11]	0.24 [2.94]	0.21 [2.49]	0.19 [2.32]	0.18 [2.23]	0.17 [2.09]	0.17 [2.07]
	(4)	0.64 [3.31]	0.69 [3.68]	0.70 [3.77]	0.77 [3.97]	0.79 [3.83]	0.16 [1.99]	0.11 [1.37]	0.09 [1.14]	0.07 [0.88]	0.14 [1.75]	0.12 [1.50]
	(5)	0.43 [2.41]	0.61 [3.84]	0.62 [3.84]	0.49 [3.09]	0.66 [3.86]	0.23 [2.53]	0.26 [2.86]	0.23 [2.53]	0.23 [2.55]	0.21 [2.28]	0.22 [2.35]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	CNS Quintiles					CNS Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	262	263	262	260	259	24	23	21	21	22	
	(2)	80	80	80	80	80	43	43	42	42	43	
	(3)	62	62	62	61	61	78	78	77	78	78	
	(4)	55	55	56	55	55	177	175	183	180	175	
(5)	52	52	52	52	52	1144	1420	1876	1438	1167		

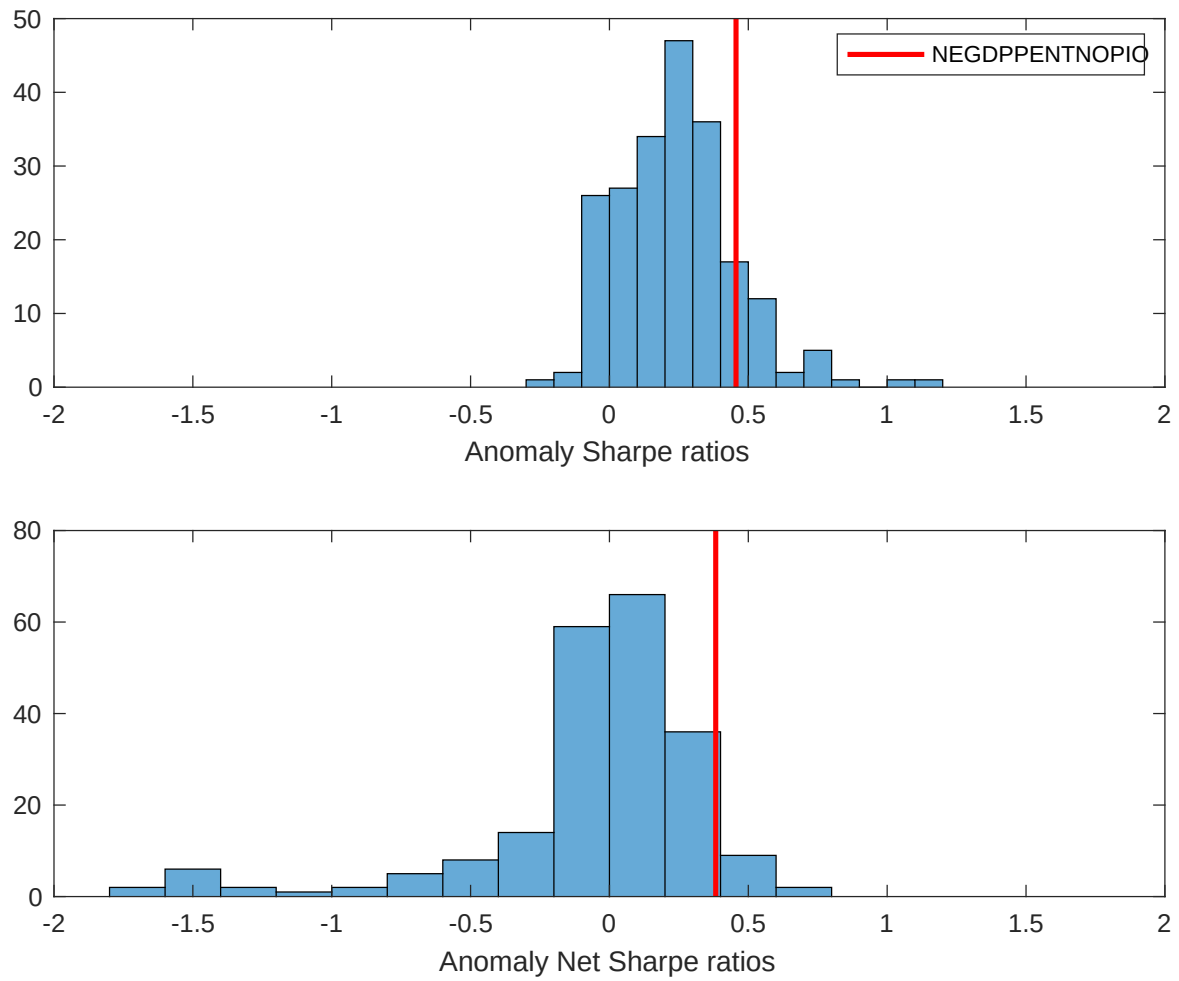


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CNS with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

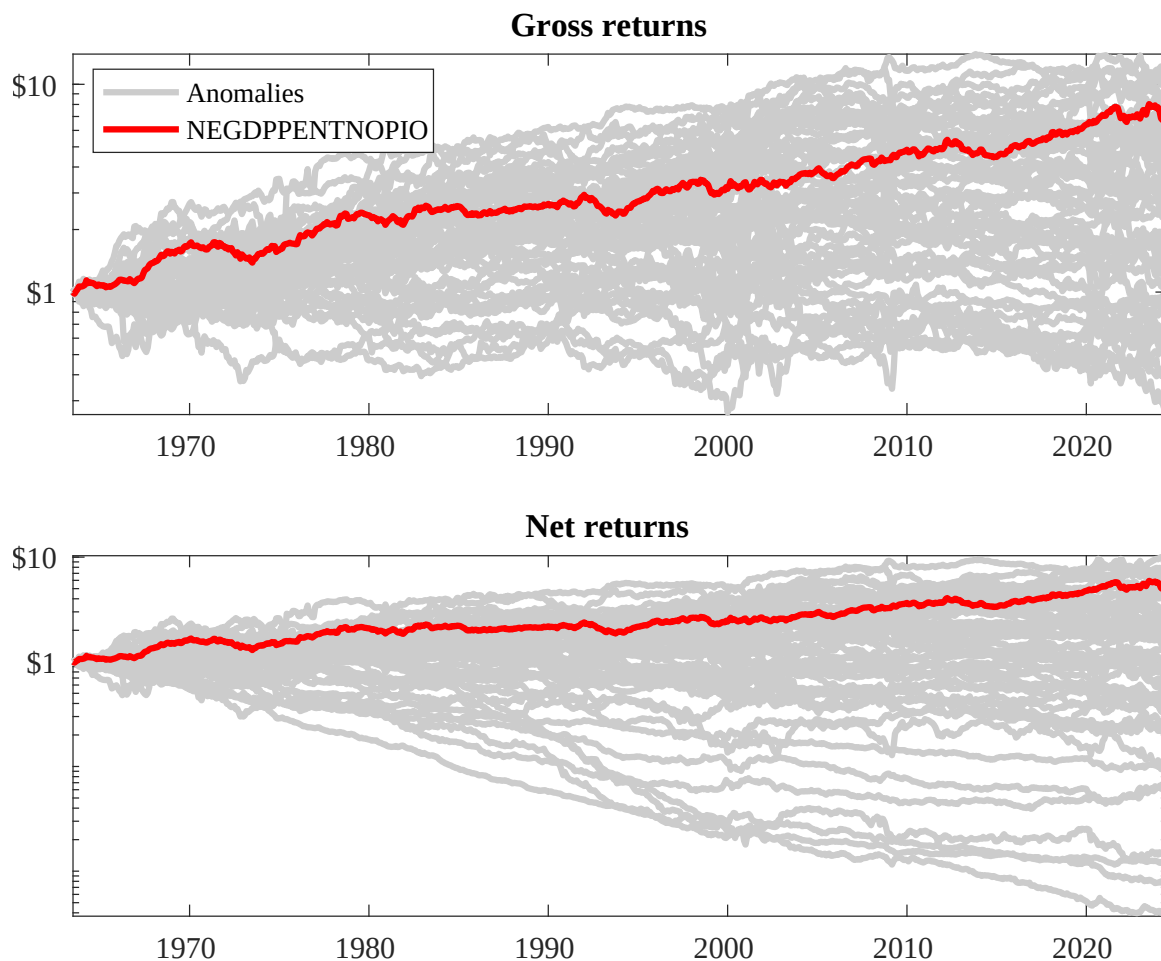
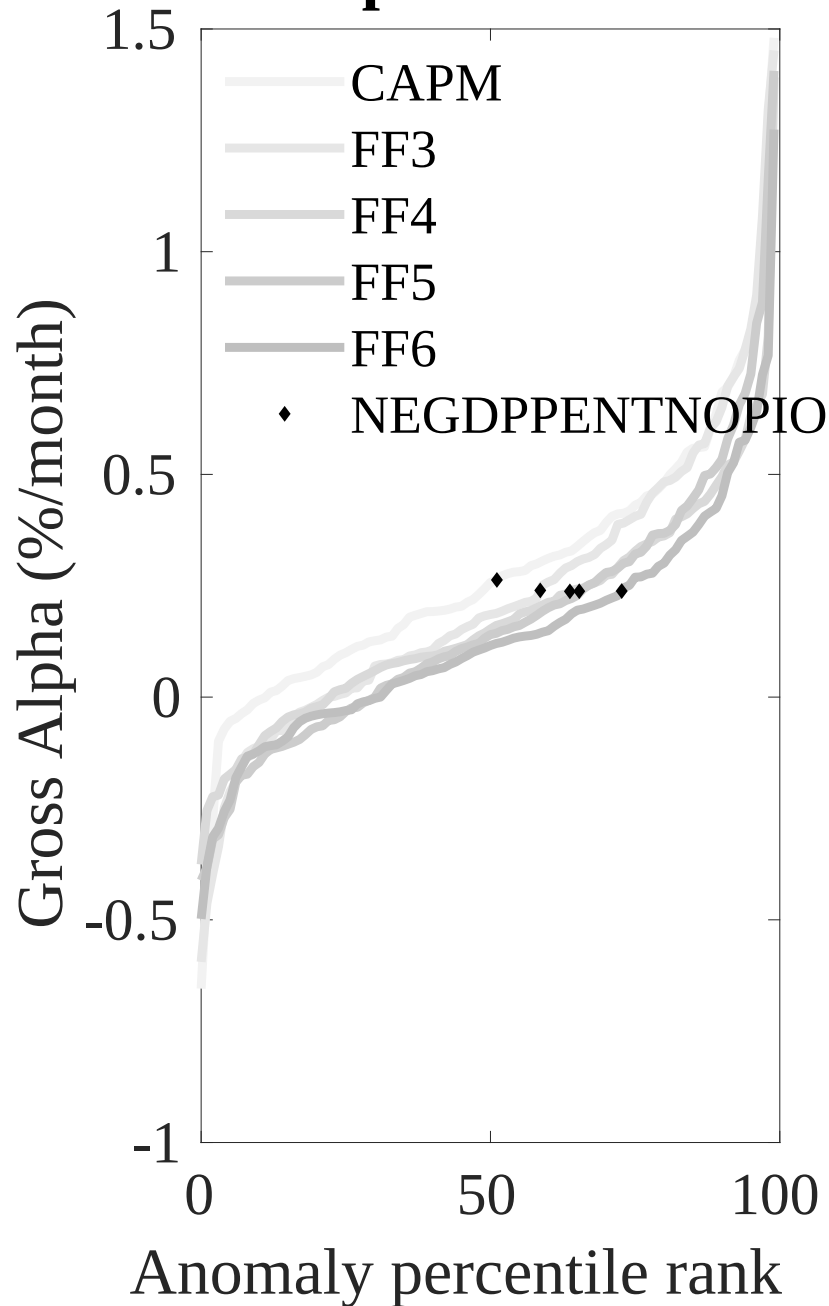


Figure 3: Dollar invested.

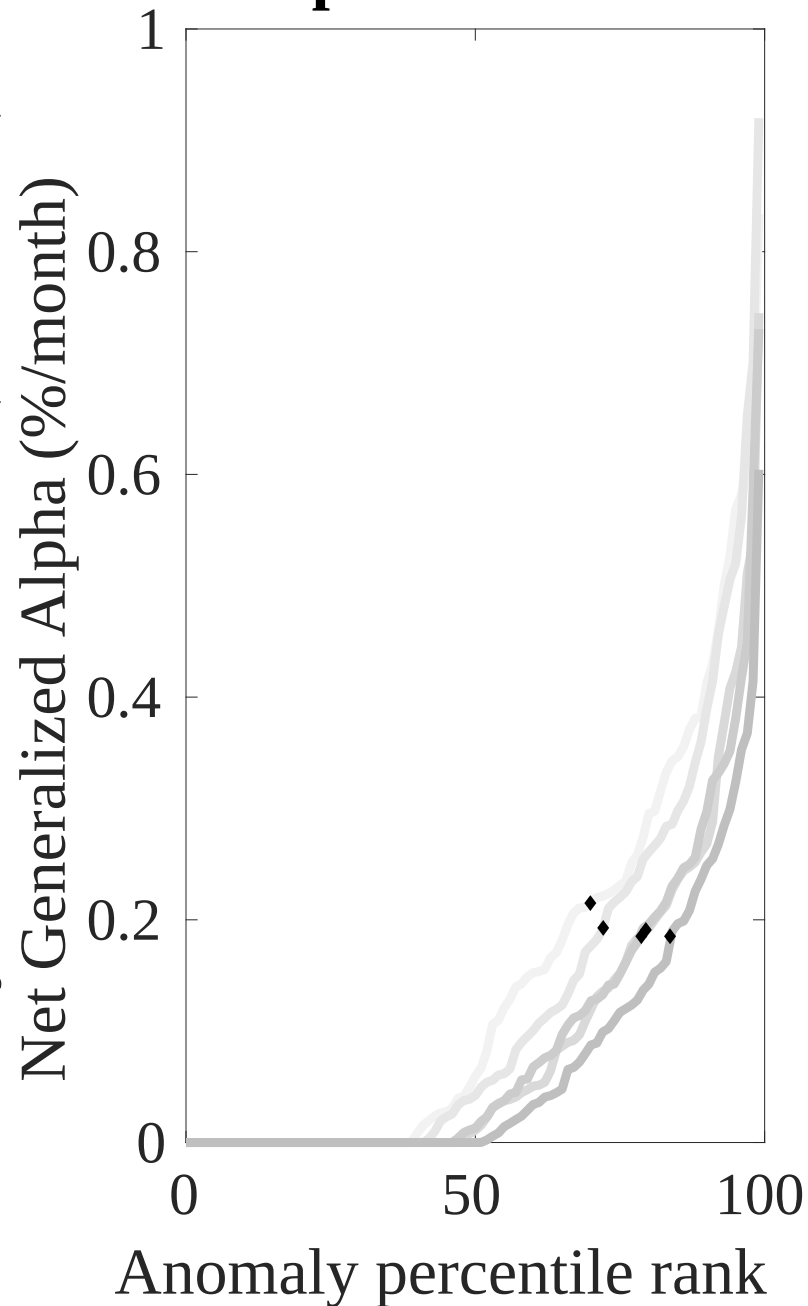
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CNS trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



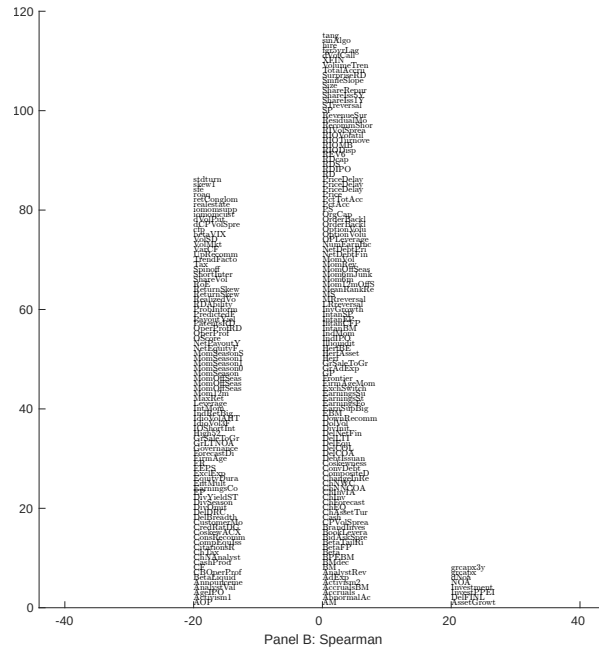
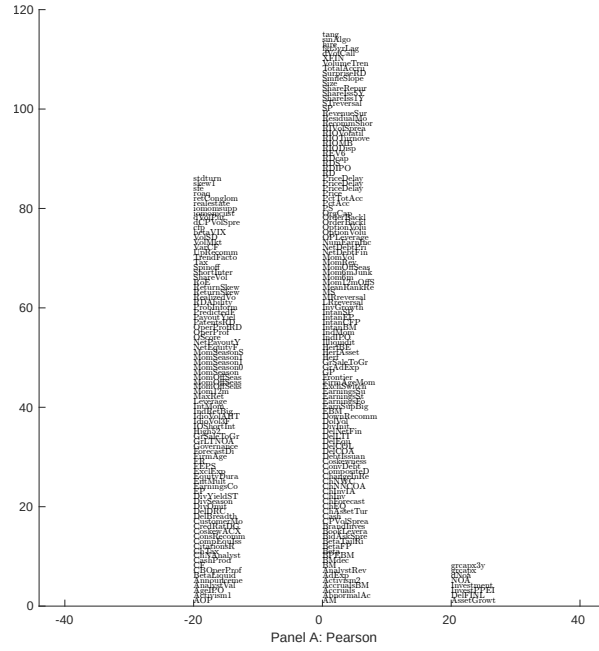


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with CNS. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

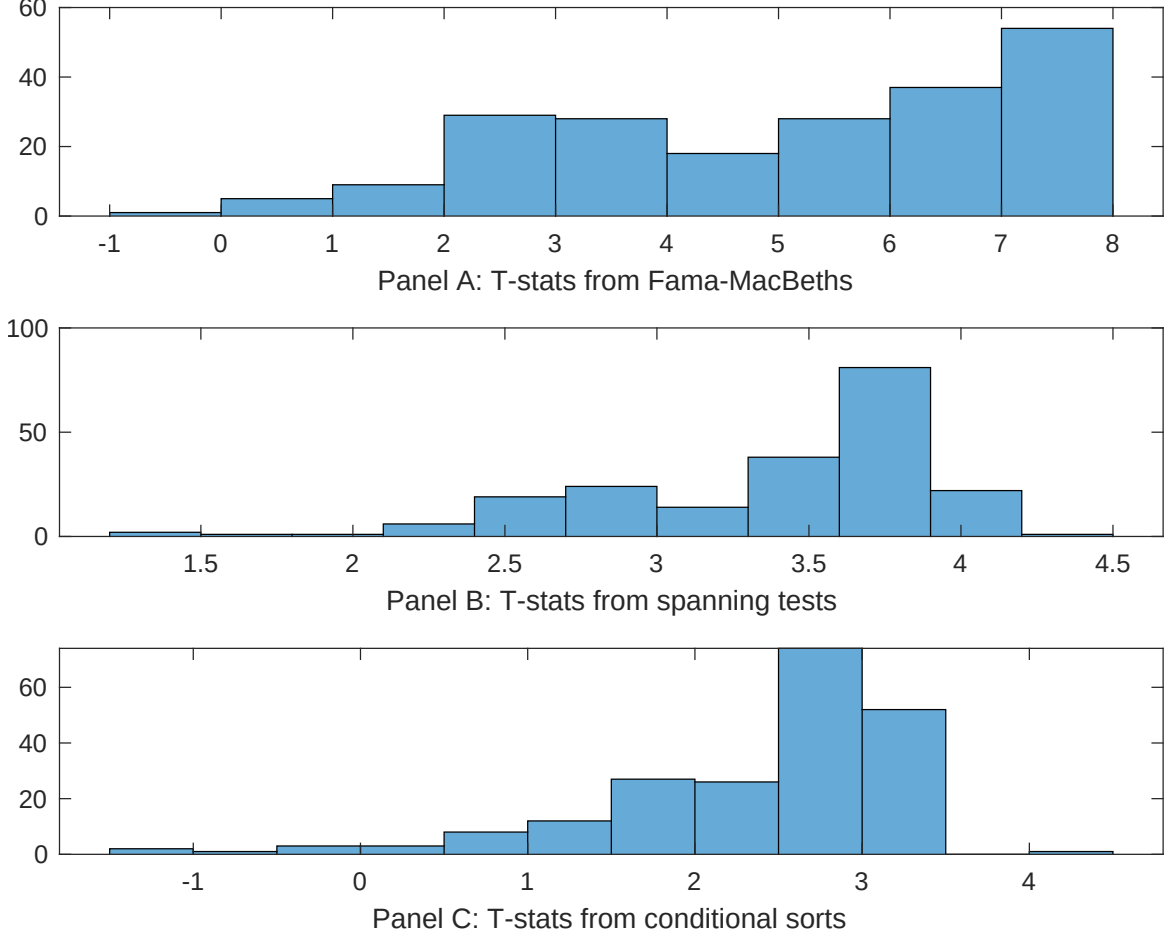


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of CNS conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CNS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CNS}CNS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CNS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CNS. Stocks are finally grouped into five CNS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CNS trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CNS. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CNS}CNS_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are change in ppe and inv/assets, change in net operating assets, Change in net financial assets, Net Operating Assets, Change in capex (three years), Inventory Growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [6.08]	0.13 [5.82]	0.12 [5.59]	0.17 [6.99]	0.14 [6.61]	0.13 [5.68]	0.15 [5.48]
CNS	0.63 [3.66]	0.70 [3.71]	0.11 [5.93]	0.94 [5.37]	0.87 [4.38]	0.90 [4.20]	0.20 [0.89]
Anomaly 1	0.16 [8.11]						0.20 [0.69]
Anomaly 2		0.13 [8.80]					1.00 [3.59]
Anomaly 3			0.89 [6.14]				-0.66 [-2.40]
Anomaly 4				0.80 [6.74]			0.52 [0.33]
Anomaly 5					0.14 [7.60]		0.83 [3.91]
Anomaly 6						0.52 [9.22]	0.67 [0.90]
# months	732	720	732	732	727	732	712
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CNS trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CNS} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are change in ppe and inv/assets, change in net operating assets, Change in net financial assets, Net Operating Assets, Change in capex (three years), Inventory Growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.24 [3.21]	0.21 [2.72]	0.19 [2.42]	0.19 [2.44]	0.26 [3.33]	0.23 [2.99]	0.20 [2.66]
Anomaly 1	28.61 [8.18]						23.26 [5.67]
Anomaly 2		14.10 [3.23]					1.46 [0.31]
Anomaly 3			19.83 [5.36]				10.62 [2.84]
Anomaly 4				17.65 [5.45]			10.85 [3.22]
Anomaly 5					6.61 [1.75]		-5.31 [-1.37]
Anomaly 6						8.42 [2.69]	-1.77 [-0.54]
mkt	-0.05 [-0.03]	0.38 [0.20]	0.64 [0.35]	-0.33 [-0.18]	0.47 [0.26]	0.19 [0.10]	0.17 [0.10]
smb	8.83 [3.42]	10.12 [3.78]	11.12 [4.19]	12.78 [4.75]	8.27 [3.00]	10.53 [3.91]	12.01 [4.42]
hml	-4.76 [-1.37]	-2.72 [-0.76]	-1.64 [-0.46]	-6.37 [-1.75]	-2.74 [-0.76]	-2.02 [-0.56]	-6.02 [-1.71]
rmw	-6.49 [-1.86]	-6.21 [-1.72]	-3.67 [-1.02]	-8.16 [-2.28]	-6.23 [-1.73]	-5.32 [-1.46]	-5.49 [-1.57]
cma	-8.98 [-1.57]	2.45 [0.40]	16.43 [3.14]	12.69 [2.44]	8.45 [1.51]	6.07 [1.03]	-0.93 [-0.14]
umd	0.11 [0.06]	-0.13 [-0.07]	-0.23 [-0.13]	-0.47 [-0.26]	-0.24 [-0.13]	-0.73 [-0.39]	-0.04 [-0.02]
# months	732	732	732	732	728	732	728
$\bar{R}^2(\%)$	12	5	7	7	4	5	13

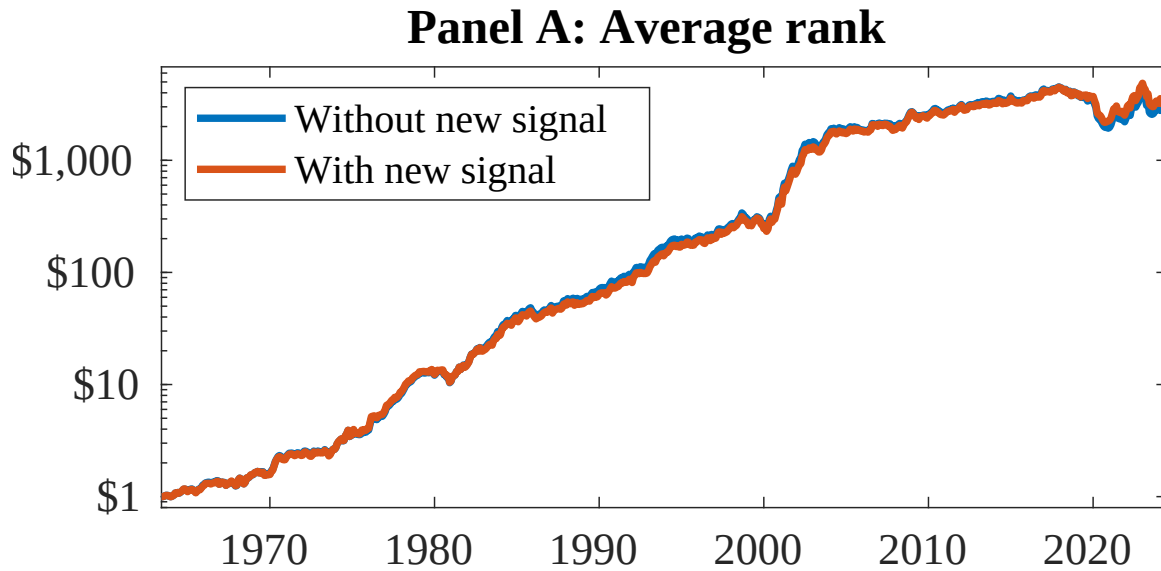


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as CNS. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

- Bansal, R. and Yaron, A. (2004a). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.
- Bansal, R. and Yaron, A. (2004b). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.
- Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3):823–866.
- Campbell, J. Y. and Cochrane, J. H. (1999). By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107(2):205–251.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Cooper, M. J., Gulen, H., and Schill, M. J. (2008). Asset growth and the cross-section of stock returns. *Journal of Finance*, 63(4):1609–1651.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.

- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Gomes, J., Kogan, L., and Zhang, L. (2003). Equilibrium cross section of returns. *Journal of Political Economy*, 111(4):693–732.
- Hirshleifer, D., Hou, K., Teoh, S. H., and Zhang, Y. (2004). Do investors overvalue firms with bloated balance sheets? *Journal of Accounting and Economics*, 38:297–331.
- Lettau, M. and Ludvigson, S. (2001). Consumption, aggregate wealth, and expected stock returns. *Journal of Finance*, 56(3):815–849.
- Li, E. X. and Zhang, L. (2010). A neoclassical interpretation of momentum. *Journal of Financial Economics*, 97(2):275–293.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.
- Xing, Y. (2008). Interpreting the value effect through the q-theory: An empirical investigation. *Review of Financial Studies*, 21(4):1767–1795.
- Zhang, L. (2005). The value premium. *Journal of Finance*, 60(1):67–103.